Long-Only Gold Trend Following Through a Bear Market

An Out-of-Sample Test on CME Gold Futures, 2011–2026

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Abstract

We test a long-only, volatility-targeted trend-following strategy on gold, designed and calibrated on broker data from 2015 onward, using fifteen years of single-provenance CME gold futures (GC) hourly data. The pre-2015 period, which contains the 2011–2015 gold bear market, is a genuine out-of-sample test. From June 2011 to December 2014 the strategy returned +4.4% (Sharpe 0.18, maximum drawdown −15.5%), while gold fell 21.4% (maximum drawdown −40.9%) and a constant long position at the same volatility target lost 13.1%. Over the full sample (June 2011 – November 2026), net of costs and implicit financing, the strategy earned 8.1% a year with a Sharpe ratio of 0.69 (95% block-bootstrap interval 0.18–1.19) and a maximum drawdown of -18.9%, against 0.32 and -45.0% for buy-and-hold. The in-sample Sharpe of about 1.08 is not reproduced. Most of the improvement over buy-and-hold is attributable to volatility targeting: the timing component's advantage over a volatility-matched constant long position (paired t = +1.00; bootstrap P(ΔSharpe ≤ 0) = 0.15) is not statistically significant, whereas its drawdown reduction is large and stable across regimes. We conclude that the strategy's robust contribution is downside control rather than return predictability.

Keywords: *Trend following, Time-series momentum, Gold, Volatility targeting, Out-of-sample testing, Futures*
JEL Classification: G11, G14, G17, Q02

1 Introduction

Backtests of trading rules are evaluated on the same data used to choose them, and their performance is therefore biased upward by an amount that grows with the number of configurations tried (Bailey & López de Prado, 2014; Harvey, Liu & Zhu, 2016). For a long-only strategy on an asset with a strong upward drift, the bias has a second source: a sample dominated by a bull market rewards any rule that is mostly long. Gold from 2015 to 2026, which rose from roughly $1,100 to above $4,000 an ounce, is such a sample.

This paper studies a strategy built under exactly these conditions. It combines exponentially weighted moving-average crossover and breakout forecasts, is long-only, concentrates its exposure when signals are strong, and targets a constant volatility. On broker data from 2015 onward it achieved a Sharpe ratio of approximately 1.02–1.10 at 4-hour frequency. Two problems prevented us from validating that figure on earlier data. First, a pre-2015 4-hour gold series was available only as a splice of two brokers' feeds, a construction that had previously produced spurious results in our research. Second, an earlier validation at daily frequency back to 2002 could not speak directly to the 4-hour implementation.

We resolve both problems with exchange data. CME Globex gold futures hourly bars from June 2010 are single-provenance, exchange-timestamped, and, once mapped to the broker's server clock, correlate at 0.986 with the instrument the strategy trades. Because the strategy's parameters were fixed using post-2015 data, every observation from June 2011 to December 2014 is out of sample. That window includes gold's largest drawdown since the early 1980s.

Our contributions are threefold. (i) We report a clean out-of-sample test of a 4-hour gold trend strategy through a 45% bear market. (ii) We decompose its performance relative to buy-and-hold into a volatility-targeting component and a timing component, and show that only the former is statistically robust. (iii) We quantify the gap between in-sample and out-of-sample Sharpe for a strategy of this kind, which is useful for setting realistic expectations.

2 Related literature

Time-series momentum, the tendency of an asset's own past returns to predict its future returns, is documented across asset classes by Moskowitz, Ooi & Pedersen (2012), and over more than a century of data by Hurst, Ooi & Pedersen (2017). Gold appears in these studies as one commodity among many; dedicated evidence is thinner, and the metal's role as a hedge or safe haven (Baur & Lucey, 2010) and its weak link to inflation over practical horizons (Erb & Harvey, 2013) make its long-run return distribution distinctive. Volatility targeting, which scales exposure inversely to recent volatility, has been shown to raise risk-adjusted returns in several asset classes (Moreira & Muir, 2017). The EWMAC and breakout forecasts used here follow the practitioner construction of Carver (2015). Our inference relies on the moving-block bootstrap (Künsch, 1989) and takes seriously the multiple-testing concerns of Bailey & López de Prado (2014) and Harvey, Liu & Zhu (2016).

3 Data

3.1 Source and construction

We use hourly OHLCV bars for CME Globex gold futures (root GC) from Databento's GLBX.MDP3 dataset, covering 7 June 2010 to 6 November 2026 (25,563 4-hour bars after resampling). The continuous series selects the most-actively-traded contract each day. Because consecutive contracts trade at different prices, we construct a back-adjusted series by setting the return to zero on every bar where the underlying contract changes, and scale open, high and low prices by the same adjustment factor as the close. The series contains 81 contract changes.

3.2 Clock alignment

The strategy operates on 4-hour bars stamped on the broker's server clock, which is New York local time plus seven hours. We convert the exchange's UTC timestamps to this clock, accounting for daylight-saving changes, and resample hourly bars into 4-hour bars with the same boundaries as the live feed.

3.3 Validation against the traded instrument

Hourly log returns of the adjusted futures series correlate with those of the broker's spot gold CFD at 0.986 at lag zero, and at −0.009 and −0.008 at lags of minus and plus one hour, confirming alignment. Daily returns correlate at 0.995. Running the strategy separately on the two data sources over their common period (2017–2026) yields daily strategy returns that correlate at 0.951.

3.4 Financing

A futures position's return already nets out the cost of carry, which for gold is close to the short-term interest rate. A spot CFD instead charges financing separately as a daily swap. We add no further financing charge, so our results correspond approximately to a CFD account that pays swap at the cash rate. Over 2017–2026 this explains most of the gap between the strategy's Sharpe on futures data (0.86) and on swap-free broker data (1.12), as short-term rates were elevated for much of 2022–2025.

4 Strategy and benchmarks

4.1 Forecasts

Let P_t denote the price at the close of 4-hour bar t, with six bars per trading day. The trend forecast averages three EWMAC rules with fast/slow spans of (16, 64), (32, 128) and (64, 256) days:

$$raw_{f,s}(t) = [\, EMA_f(P)_t - EMA_s(P)_t \,] \,/\, (\, P_t\, \sigma_t \,) \tag{1}$$

where σ_t is the exponentially weighted standard deviation of returns (span 36 bars). Each rule is normalised by the expanding mean of its own absolute value, using only past data, so that an average-strength signal has magnitude one. The three are averaged, scaled by 1.3, and capped at ±2. The breakout forecast averages three rules with look-backs N of 10, 20 and 40 days:

$$raw_N(t) = EMA_{N/4}(\, 4 \cdot [\, P_t - mid_N(t) \,] \,/\, range_N(t) \,) \tag{2}$$

where mid and range are the midpoint and width of the rolling N-day high–low channel. The same normalisation applies, with scale 1.2 and cap ±2.

4.2 Long-only concentration

With trend forecast E_t and breakout forecast B_t, and the concentration operator $c(x)$ = normalised (max(x, 0))$^{1.5}$, the final forecast is

$$F_t = ½ [\ 0.8\ c(\ max(E_t^+, B_t^+)\ )\ + 0.15\ ] + ½ [\ 0.8\ c(B_t) + 0.15\ ],\ \ clipped\ to\ [0,\ 2]. \tag{3}$$

The constant 0.15 is a resting long position held in the absence of any trend. The exponent 1.5 concentrates exposure in strong-signal states.

4.3 Position sizing and execution

The target position, as a multiple of equity, is $w_t = F_t \cdot \tau / \hat{\sigma}_t$, with τ = 10% annualised and $\hat{\sigma}_t$ the exponentially weighted volatility of 4-hour returns (half-life 42 bars, annualised with 1,512 bars per year), capped at 8. A no-trade buffer of 10% of $\tau/\hat{\sigma}_t$ suppresses small rebalances. Positions decided at the close of bar t earn the return of bar t+1. Transaction costs are 1 basis point of notional per unit of turnover; Section 6 varies this from 0 to 10 bp.

4.4 Benchmarks and sample split

We compare against (a) gold buy-and-hold on the adjusted futures series, and (b) a constant long position with $F_t \equiv 1$ passed through the identical volatility-targeting engine. Benchmark (b) isolates the contribution of timing from that of volatility scaling. Because the slowest EWMAC rule requires roughly one year of data, all series begin on 1 June 2011. The out-of-sample period is June 2011 – December 2014; the design period is 2015 onward.

5 Results

5.1 Out-of-sample and sub-period performance

Period / strategy	Total return	Sharpe	Max drawdown
Out-of-sample, Jun 2011 – Dec 2014			
Champion	**+4.4%**	**+0.18**	**-15.5%**
Constant long, same vol	-13.1%	-0.34	-28.0%
Gold buy & hold	-21.4%	-0.24	-40.9%
Gold bear market, Sep 2011 – Dec 2015			
Champion	**-5.6%**	**-0.12**	**-15.5%**
Constant long, same vol	-22.4%	-0.56	-29.0%
Gold buy & hold	-42.2%	-0.58	-45.0%
Design era, Jan 2015 – Nov 2026			
Champion	**+171.3%**	**+0.83**	**-18.9%**
Constant long, same vol	+118.3%	+0.83	-12.9%
Gold buy & hold	+142.8%	+0.53	-30.1%
Full sample, Jun 2011 – Nov 2026			
Champion	**+183.2%**	**+0.69**	**-18.9%**
Constant long, same vol	+89.6%	+0.52	-29.0%
Gold buy & hold	+90.7%	+0.32	-45.0%

Table 1. Performance by period. Sharpe ratios are annualised from daily returns. Strategies are scaled to a 10% annual volatility target; gold buy-and-hold is unscaled. The out-of-sample period precedes all data used to design the strategy.

Table 1 reports the central result. In the out-of-sample period the strategy returned +4.4% while gold lost 21.4% and the volatility-matched constant long lost 13.1%. Its maximum drawdown (−15.5%) was 38% of gold's. Over the bear market proper (September 2011 – December 2015) the strategy lost 5.6% against 42.2% for gold. In the design period it delivered a Sharpe of 0.83, equal to the constant long's, with a larger drawdown. Over the full sample its Sharpe of 0.69 exceeds both benchmarks, at well under half of gold's maximum drawdown.

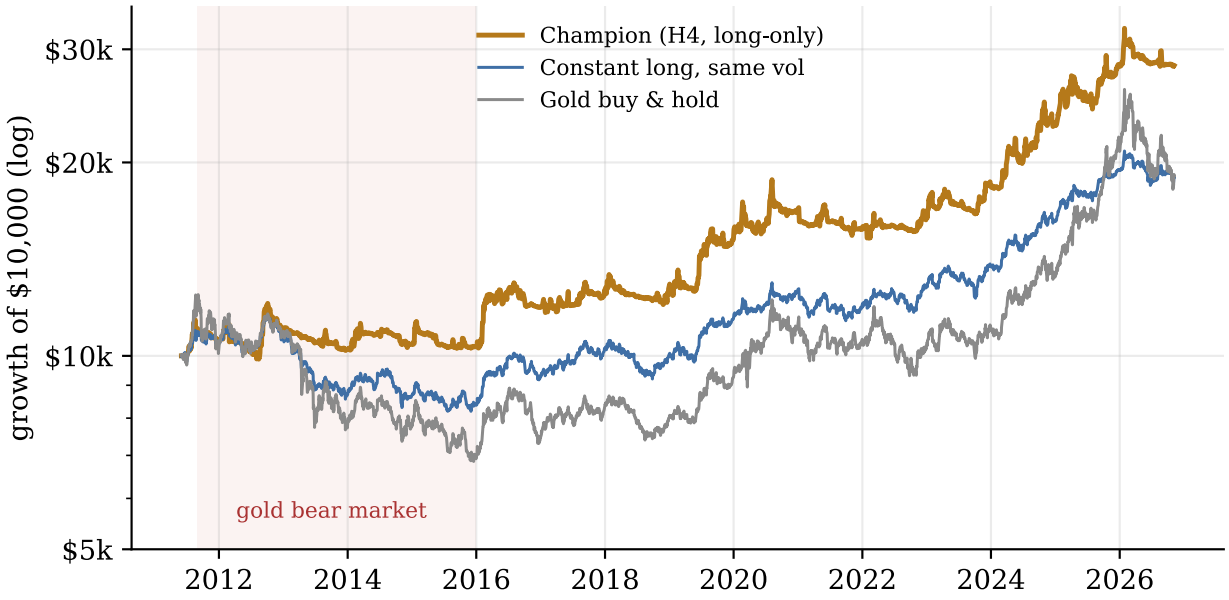


Figure 1. Growth of $10,000, June 2011 – November 2026, log scale. Shading marks the 2011–2015 gold bear market.

5.2 Full-sample risk and return

	Champion	Constant long	Gold B&H
Total return	+183%	+90%	+91%
CAGR	+8.1%	+4.9%	+3.9%
Annualised volatility	12.3%	10.2%	16.3%
Sharpe ratio	0.69	0.52	0.32
Sharpe 95% CI (block bootstrap)	[0.18, 1.19]	[0.02, 1.04]	[-0.14, 0.78]
Sortino ratio	0.85	0.73	0.42
Maximum drawdown	-18.9%	-29.0%	-45.0%
Calmar (CAGR / max DD)	0.43	0.17	0.09
Longest time under water	3.3 yrs	6.5 yrs	13.7 yrs
Positive months	44%	52%	49%
Best / worst month	+15.9% / -5.9%	+7.9% / -6.0%	+13.0% / -12.0%
Skewness of monthly returns	+1.40	+0.10	+0.10

Table 2. Full-sample statistics, June 2011 – November 2026, net of 1 bp costs. Confidence intervals: moving-block bootstrap with 21-day blocks and 2,000 resamples.

The strategy's return profile is characteristic of trend following: only 44% of months are positive, but monthly returns are strongly right-skewed (skewness +1.40, against +0.10 for the constant long), with a best month of +15.9% and a worst of -5.9%. Realised volatility (12.3%) exceeds the 10% target because the forecast can reach 2.

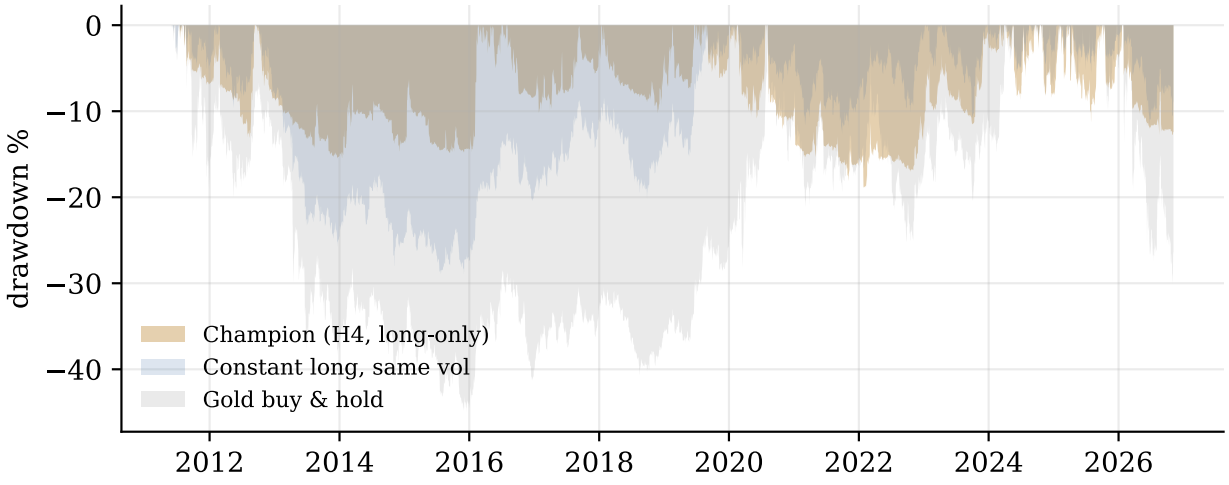


Figure 2. Drawdown from the running maximum.

5.3 Behaviour in the bear market

Between September 2011 and December 2015 the forecast averaged 0.56, sat at the resting level (below 0.2) on 52% of bars, and reached strong-trend territory (≥ 1.5) on 14%. Average exposure fell to 0.40× equity, against 0.76× from 2016 onward. Figure 3 shows the mechanism: each counter-trend rally produced a short burst of exposure, which was unwound within weeks when the breakout failed. The strategy's bear-market losses are the sum of these false starts, each small relative to the decline it avoided.

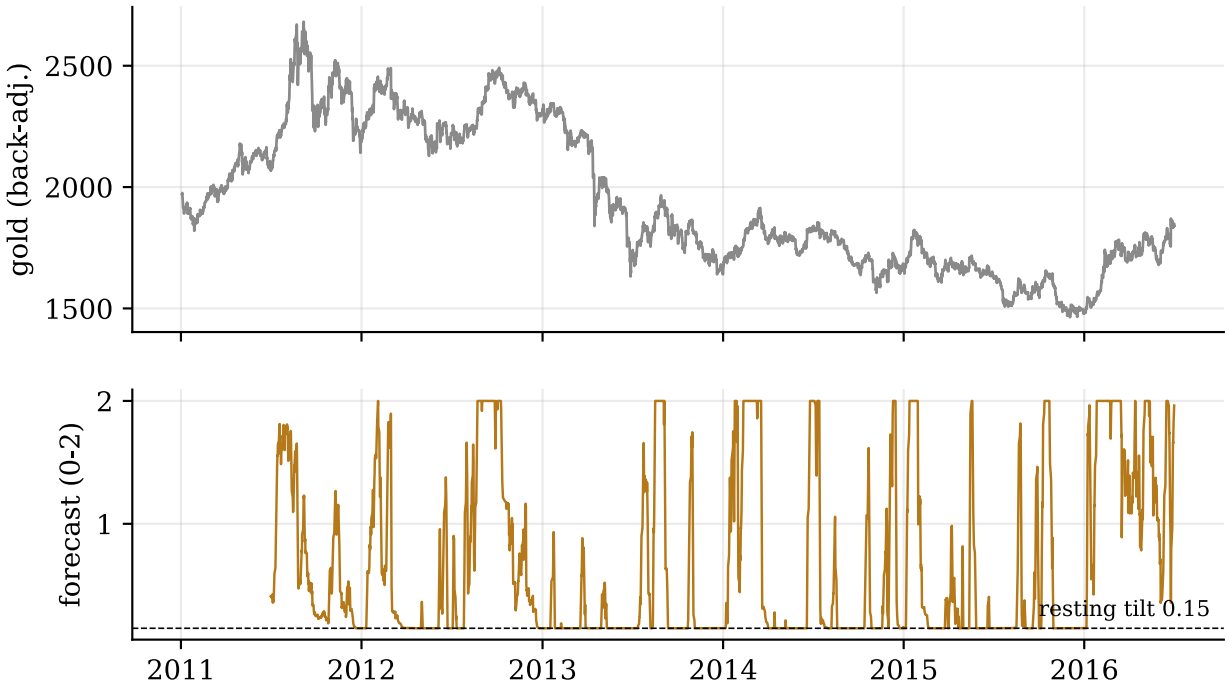


Figure 3. Adjusted gold futures price (top) and the strategy forecast F_t (bottom), 2011–2016. The dashed line marks the 0.15 resting level.

5.4 Timing versus volatility scaling

To isolate timing, we compare the strategy's daily returns with those of the constant long rescaled to the same realised volatility. The mean difference has a paired *t*-statistic of +1.00 over the full sample (+1.29 out of sample, +1.39 in the bear market, +0.34 in the design period). A block bootstrap of the Sharpe-ratio difference gives a 95% interval of [-0.14, +0.47] and P(ΔSharpe ≤ 0) = 0.15. Against gold buy-and-hold the corresponding interval is [-0.08, +0.60] with P(ΔSharpe ≤ 0) = 0.055. The evidence that the strategy improves on owning gold is therefore considerably stronger than the evidence that its timing improves on a volatility-scaled constant position. Timing contributed mainly in the bear market, where it reduced drawdown; in the design period it added nothing to the Sharpe ratio.

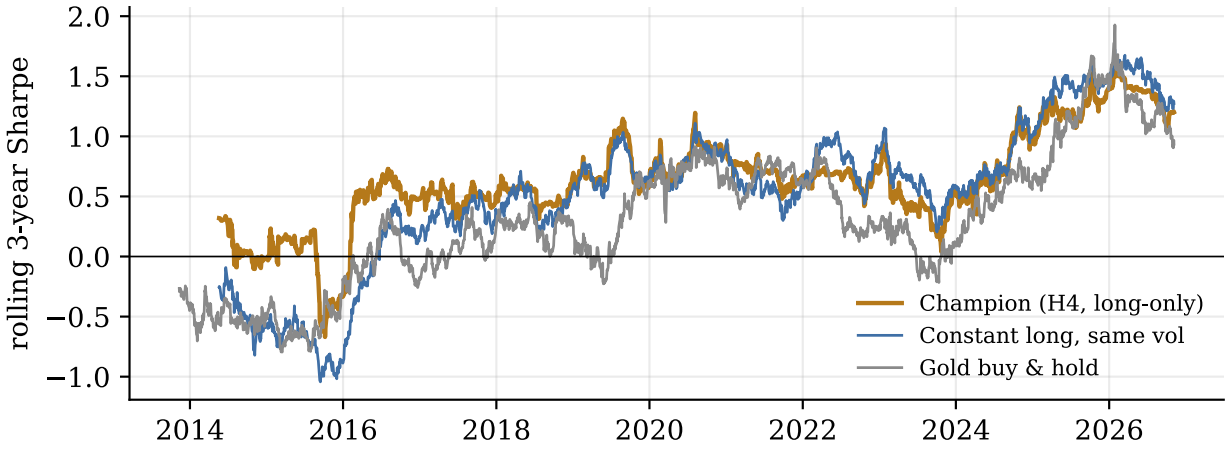


Figure 4. Rolling three-year Sharpe ratio (756 trading days). It was negative for windows ending in 2014–2015, between roughly 0.5 and 0.9 for windows ending 2016–2023, and above 1 only for windows ending in 2025–2026. The two longest drawdown spells lasted 3.3 years (October 2012 – February 2016) and 2.9 years (August 2020 – March 2024).

6 Robustness

One-way cost	Sharpe
0 bp	0.70
1 bp	0.69
2 bp	0.68
3 bp	0.67
5 bp	0.64
10 bp	0.58

Table 3. Full-sample Sharpe ratio as a function of the transaction cost per unit of turnover. Average turnover is 28× equity a year.

Transaction costs. Table 3 shows that the result is insensitive to execution costs: at ten times the baseline cost the Sharpe declines by about 0.11. Financing, which the futures data embed, is the more material cost for a long-only gold strategy.

Data provenance. On the common period 2017–2026 the strategy's returns on futures and on broker data correlate at 0.951. The difference in Sharpe (0.86 versus 1.12) is consistent with the financing embedded in futures and absent from the swap-free broker backtest.

Bar frequency and longer history. In a companion test, a daily-frequency version of the same rules, with all speeds expressed in days, was run on continuous gold futures from 2002. Over 2002–2026 it achieved a Sharpe of 0.64 with a maximum drawdown of −17.3%, against 0.69 and −44.4% for buy-and-hold and 0.75 and −32.6% for a volatility-matched constant long, and it lost 11.2% in the 2011–2015 bear market while gold lost 42.0%. The 4-hour and daily results agree: similar risk-adjusted return to a volatility-scaled long position, with roughly half its drawdown.

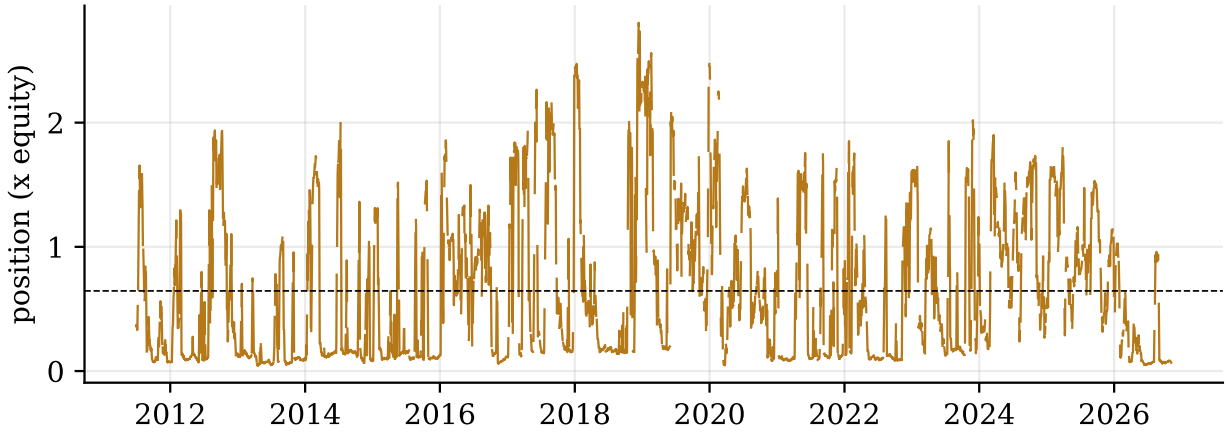


Figure 5. Daily average position as a multiple of equity. Dashed line: sample mean (0.65×).

7 Limitations

First, fifteen years contains few independent trend episodes, so the Sharpe ratio's confidence interval ([0.18, 1.19]) is wide. Second, although the parameters were fixed on post-2015 data, the strategy family was selected after extensive research on that same data, and this selection is not fully captured by the out-of-sample split. Third, back-adjusted futures differ from spot exposure through the term structure; our validation suggests the difference is small at the frequencies studied. Fourth, we model costs proportionally and ignore market impact, which is negligible at retail size in gold futures but not at institutional scale. Fifth, the strategy is single-asset; diversification across trend-following markets would change its risk profile substantially.

8 Conclusion

A long-only, volatility-targeted gold trend strategy designed on 2015–2026 data survives a clean out-of-sample test through the 2011–2015 bear market, preserving capital while gold lost a fifth of its value. Over fifteen years its Sharpe ratio is 0.69 with financing included, well below the in-sample estimate near 1.1 but roughly double that of buy-and-hold, and its maximum drawdown is less than half of gold's. Its advantage over a volatility-matched constant long position is directionally consistent but not statistically significant. The robust value of the strategy lies in drawdown control: it delivers gold's long-run return

with materially smaller and shorter losses. For practitioners, the gap between the in-sample and out-of-sample Sharpe ratios is a reminder to size and evaluate such strategies on out-of-sample expectations.

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Appendix A Annual returns

Year	Champion	Constant long	Gold B&H
2011	+6.0%	+4.4%	+2.2%
2012	+4.2%	+1.8%	+6.2%
2013	-7.3%	-18.1%	-28.3%
2014	+2.0%	-0.2%	+1.0%
2015	-1.1%	-3.6%	-12.0%
2016	+15.4%	+10.7%	+7.7%
2017	+6.2%	+10.8%	+11.8%
2018	+0.5%	-2.6%	-3.8%
2019	+19.3%	+14.0%	+15.6%
2020	+11.5%	+11.2%	+20.9%
2021	-6.4%	-5.4%	-5.2%
2022	+4.2%	+7.0%	-2.6%
2023	+11.8%	+7.6%	+7.2%
2024	+24.1%	+17.7%	+20.6%
2025	+23.3%	+19.4%	+55.7%
2026*	+0.3%	-2.4%	-8.3%

Table A1. Calendar-year returns in percent. * 2011 from 1 June; 2026 to 6 November. The strategy was positive in 81% of years and outperformed gold in 69%.

Replication. Data: Databento GLBX.MDP3, ohlcv-1h, symbol GC.v.0. Code: `scripts/fetch_databento_ohlcv.py`, `scripts/v5_champion_h4_cme.py`, `scripts/v5_champion_paper_assets.py`; signal definition in `src/v5/xau_dual_signals.py`.

Disclaimer. This paper reports simulated historical results. It is not investment advice. Past performance, simulated or actual, does not guarantee future results.